Solution

Let $a(t)$ be the distance of car A north of P at time t, and $b(t)$ the distance of car B east of P at time t, and let $c(t)$ be the distance from car A to car B at time t. By the Pythagorean Theorem, $c(t)^2 = a(t)^2 + b(t)^2$.

Taking derivatives, we get $2c(t)c'(t) = 2a(t)a'(t) + 2b(t)b'(t)$.

$$\text{So, } c' = \frac{aa' + bb'}{c} = \frac{aa' + bb'}{\sqrt{a^2 + b^2}}$$

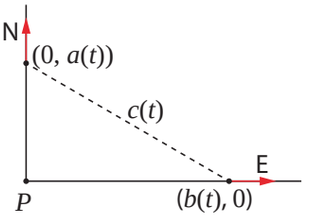


Fig.7.6

Substituting known values, we get

$$c' = \frac{(10\times 80) + (15\times 100)}{\sqrt{10^2 + 15^2}} = \frac{460}{\sqrt{13}} \approx 127.6 \text{ km/hr}$$

at the time of intersect.

EXERCISE 7.1

1. A particle moves along a straight line in such a way that after t seconds its distance from the origin is $s = 2t^2 + 3t$ metres.
   (i) Find the average velocity between $t = 3$ and $t = 6$ seconds.
   (ii) Find the instantaneous velocities at $t = 3$ and $t = 6$ seconds.
2. A camera is accidentally knocked off an edge of a cliff 400 ft high. The camera falls a distance of $s = 16t^2$ in t seconds.
   (i) How long does the camera fall before it hits the ground?
   (ii) What is the average velocity with which the camera falls during the last 2 seconds?
   (iii) What is the instantaneous velocity of the camera when it hits the ground?
3. A particle moves along a line according to the law $s(t) = 2t^3 - 9t^2 + 12t - 4$, where $t \geq 0$.
   (i) At what times the particle changes direction?
   (ii) Find the total distance travelled by the particle in the first 4 seconds.
   (iii) Find the particle's acceleration each time the velocity is zero.
4. If the volume of a cube of side length x is $v = x^3$. Find the rate of change of the volume with respect to x when $x = 5$ units.
5. If the mass $m(x)$ (in kilograms) of a thin rod of length x (in metres) is given by, $m(x) = \sqrt{3x}$ then what is the rate of change of mass with respect to the length when it is $x = 3$ and $x = 27$ metres.
6. A stone is dropped into a pond causing ripples in the form of concentric circles. The radius r of the outer ripple is increasing at a constant rate at 2 cm per second. When the radius is 5 cm find the rate of changing of the total area of the disturbed water?
7. A beacon makes one revolution every 10 seconds. It is located on a ship which is anchored 5 km from a straight shore line. How fast is the beam moving along the shore line when it makes an angle of $45°$ with the shore?
8. A conical water tank with vertex down of 12 metres height has a radius of 5 metres at the top. If water flows into the tank at a rate 10 cubic m/min, how fast is the depth of the water increases when the water is 8 metres deep?
9. A ladder 17 metre long is leaning against the wall. The base of the ladder is pulled away from the wall at a rate of 5 m/s. When the base of the ladder is 8 metres from the wall,
   (i) how fast is the top of the ladder moving down the wall?
   (ii) at what rate, the area of the triangle formed by the ladder, wall, and the floor, is changing?

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